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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I R.Rupasree (192H72122) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R.RupaSree

20+29+19+19+10 = 97

RUBRICS FOR CALCULATION

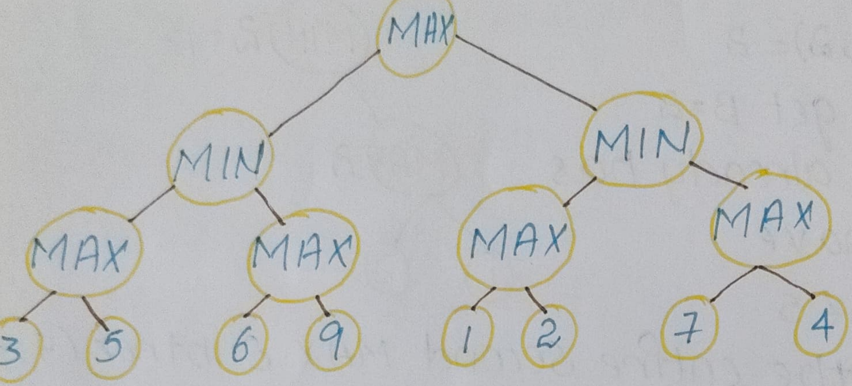
Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

20+29+19+19+10

97

Q) Alpha-Beta Pruning:-

Evaluate the displayed game tree from left to right using Alpha-Beta pruning. show alpha and beta values, Identify pruned nodes and determine the final minimax value.

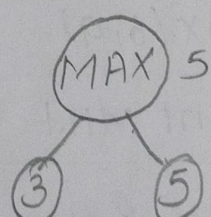


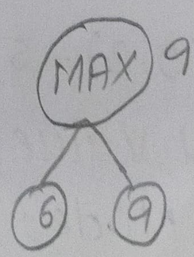
S.No	Node	Alpha	Beta	Action / Pruning
1				
2				
3				
4				
5				

Sol:- Alpha-Beta pruning:- The tree is evaluated from left to right.

Step-1: Left Subtree:-

- First Max node:- $MAX(3,5) = 5$
- MIN Node gets value 5
- Second MAX Node:- $MAX(6,9) = 9$
- MAX already gets 6, and MIN has $\beta = 5$. the Value 6 is not enough to make $\alpha \geq \beta$
- Therefore, 9 is pruned.

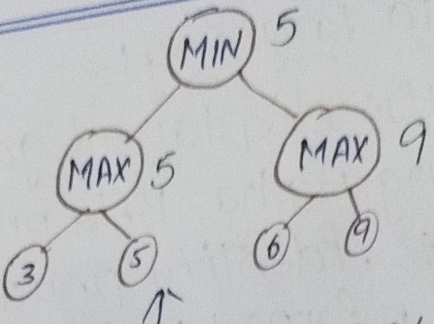




so, left MIN:

MIN (5,6) = 5

Root MAX gets $\alpha = 5$



$\alpha = 5 \rightarrow \beta \leq \alpha = 2 \leq 5$

Step2: Right subtree

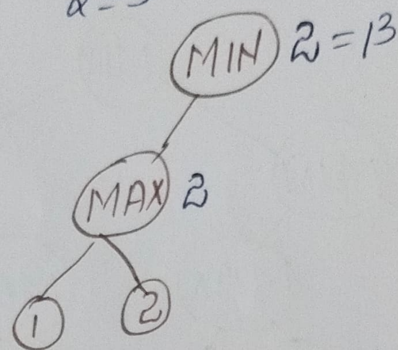
•) First MAX Node:

MAX (1,2) = 2

•) Right MIN get $\beta = 2$

•) Since root already has $\alpha = 5$ we have

$\beta \leq \alpha \rightarrow 2 \leq 5$



Therefore, the entire second MAX subtree (7,4) is pruned.

Alpha-Beta Table:-

S.NO	NODE	α	β	Action/pruning
1	Root MAX	5	∞	Gets left values 5
2	Left MIN	$-\infty$	5	Gets minimum=5
3	MAX (3,5)	5	∞	Value = 5
4	MAX (6,9)	6	5	9 pruned
5	Right MIN	5	2	(7,4) pruned

Final Minimax Value:-

MAX (5,2) = 5

So, MIN Value = 5

pruned Nodes: 9, 7 and 4.

Q) Evaluation Function:-

An AI game-playing agent uses $E(S) = 2M + 3K + 5Q + P$

For S1 : $M = 4, K = 2, Q = 1, P = 5$

For S2 : $M = 3, K = 5, Q = 2, P = 4$

Calculate both evaluation score and determine which state MAX should select.

→ Given:

$$E(S) = 2M + 3K + 5Q + P$$

S1 : $M = 4, K = 2, Q = 1, P = 5$

S2 : $M = 3, K = 5, Q = 2, P = 4$

State S1:

$M = 4, K = 2, Q = 1, P = 5$

$$E(S1) = 2(4) + 3(2) + 5(1) + 5$$
$$= 8 + 6 + 5 + 5$$
$$= 24$$

$$\boxed{E(S1) = 24}$$

State S2:

$M = 3, K = 5, Q = 2, P = 4$

$$E(S2) = 2(3) + 3(5) + 5(2) + 4$$
$$= 6 + 15 + 10 + 4$$
$$= 35$$

$$\boxed{E(S2) = 35}$$

Comparison:-

S1 : 24

S2 : 35

Since MAX selects the state with higher evaluation value. MAX should select S2

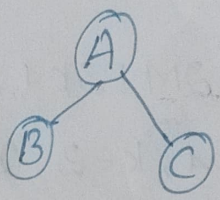
3) Search with Partial Information:-

A robot must travel from A to G. Initially it knows only A→B and A→C. On reaching B it discovers B→D and later D→G. Show how the search space changes as new information is obtained.

↳ Initially, the robots knows only:

A → B

A → C



So, the initial space search :⤴

The robot doesn't initially know where these path lead.

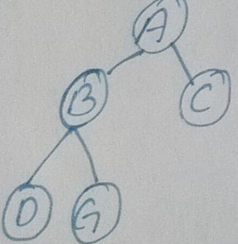
After Reaching B

The robot discovers:

B → D

B → G

The search space becomes:



Now, the robot has found direct path:

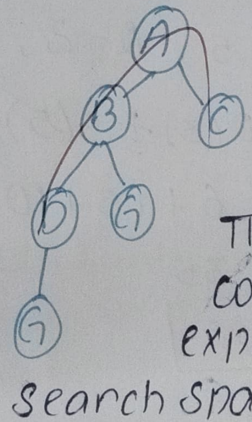
A → B → G

After Reaching D

The robot later discovers:

D → G

The search space becomes:



Thus, the robot continuously expands its search space as it receives new information

Conclusion:-

This is example of search with partial information, where complete search space is known initially. The robot update its model while exploring & can choose path A → B → G (Goal)